

Eswar Adapa

Data Scientist

(+91) 6309925636 | Mumbai, India | eswaradapa90@gmail.com
[GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 2+ years of experience in software engineering, system design, and end-to-end delivery of AI-powered systems. Builds scalable, production-grade solutions in Big Data Engineering, Machine Learning, Knowledge Graphs, and Generative AI. Scaled a POI pipeline from 23M to 220M records, architected an Agentic AI platform with RAG serving, and received Best Employee Award (February 2024 to May 2025).

SKILLS

Languages	Python, C, SQL
ML & AI	Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Transformers (DistilBERT), Recommender Systems, OCR (Tesseract)
LLM & Agentic AI	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Agentic AI, Prompt Engineering, Guardrails, Memory and Context Manage
Big Data	PySpark, Apache Sedona, Hadoop, Delta Lake, Airflow, Spatial Joins, Parquet
Data & Backend	pandas, FastAPI, REST APIs, Pydantic, RabbitMQ, WebSockets
Databases	MongoDB, Redis, MySQL, Qdrant, FAISS, ChromaDB
Graph & Spatial	Neo4j, Graph Data Science (GDS), Knowledge Graphs, Graph RAG, H3 Indexing
Cloud & DevOps	Amazon Web Services (AWS) (EC2, S3, EKS, SageMaker, Bedrock), Docker, Kubernetes, Jenkins, CI/CD, MLOps
Observability	Langfuse, Datadog, New Relic, Loguru

EXPERIENCE

Data Scientist / AI Engineer, Infinite Analytics	February 2024 to Present
- POI Ingestion and Data Scaling - Consolidated POI crawling, ingestion, and validation pipelines using RabbitMQ, Redis, MongoDB, and FastAPI, expanding coverage from 23M POIs across 4 countries to 220M POIs across 38 countries. - Raised POI-type mapping accuracy from 83% to 86.7% by replacing a LightGBM cascade (51 models) with a unified DistilBERT transformer; trained on 2.5M POIs with 0.005s inference latency. - Reduced geospatial error rate from 18% to 14% through plus-code mappings and triangular normalization. - Accelerated Point-to-Polygon mapping via Apache Sedona, cutting runtime from 1 week to 2.5 hours using a partition-aware pre-join strategy and Iceberg Metastore. - Sherlock AI: Multi-Agent Intelligence and Decision Platform - Migrated Sherlock AI from a monolithic LangChain agent to an Agentic AI system using LangGraph with 7 specialists, tool/function calling, memory and context management, and MongoDB persistent state; achieved 10x throughput, 31% lower token cost, and response times from 8-12s to 3-5s. - Designed plan-aware guardrails with action classification, human-validated insights, Qdrant semantic schema retrieval for grounding, and a 3-phase onboarding workflow with checkpoint-based recovery. - Engineered a production RAG pipeline with document ingestion, chunking, vector search, reranking, and response synthesis across Qdrant, FAISS, and ChromaDB; created an LLM factory spanning 5 providers with config-driven routing. - POI Knowledge Graph (Neo4j + GDS) - Constructed a 15-node-type, 14-relationship Knowledge Graph over 83M+ POIs, enabling Graph RAG and reasoning over structured and unstructured enterprise data. - Generated multi-modal embeddings via Node2Vec, FastRP, HashGNN, and GraphSAGE with KNN similarity, achieving 8x speedup on person data and 6-15x speedup on POI ingestion. - Owned production systems end-to-end: monitoring via Langfuse, Datadog, and New Relic; deployed via Docker, Jenkins CI/CD, and Kubernetes.	
Research Intern, DRDO	May 2023 to August 2023
- sEMG Signal Analysis and Gait Modeling - Trained deep learning models on IMU and sEMG data to classify 7 gait phases with 92% accuracy; refined preprocessing to yield a 20% signal quality increase. - Applied Symbolic Transfer Entropy to quantify muscle interactions, identifying a 25% rise in inter-muscle information transfer correlated with walking speed.	

PUBLICATIONS

Fatigue Classification and Onset Estimation Using sEMG Signals During Strength Training August 2022 to April 2023
APSIPA'23 - IEEE

- Presented at APSIPA'23 conference, Taiwan (IEEE); achieved **86% classification accuracy** using ML models with Majority Windowing and a **12% estimation error**.

EDUCATION

Indian Institute of Information Technology, Sricity December 2020 to April 2024
B.Tech.(Honors), Electronics and Communication Engineering

- CGPA: 8.02

ACHIEVEMENTS

Best Employee Award February 2024 to May 2025

- Recognized for leading 2 of 3 high-impact projects that expanded business coverage from 4 to 38 countries, spanning Data Engineering, Knowledge Graph development, and GenAI integration.

Academic Grand Challenge on Climate Change, Nasscom and Telangana Government February 2023 to March 2023

- Ranked among top 40 teams out of 7,000+ participants; built a predictive framework for heatwave events and Air Quality Index in Tier-2 cities of Telangana.

POSITION OF RESPONSIBILITY

Internship Manager - BITS Pilani Collaboration June 2025 to July 2025

- Served as Scrum Master for 5 BITS Pilani interns, directing sprint planning, task allocation, and mentoring on technical practices.